

Enhancing Ad Effectiveness: An Experiment on Specific vs. General Fundraising Appeals

Sarah Ki, Marisa Lenci, Jaekwang Shin, Jonathan Ho

4/16/2025

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Experiment Overview

Background

We partnered with the South Yuba River Citizens League (SYRCL), a nonprofit organization in Northern California working on efforts to protect and restore the waters, forests, and communities of the Yuba River watershed.

Research Question

Does providing potential donors with specific, quantifiable outcomes of an organization's initiatives increase donor engagement compared to providing only general information about the organization's mission?

Hypothesis

Based on previous research (Cryder et al., 2013), we expect the treatment will increase donation behavior relative to control.

Experiment Design: Treatment

Control

 **SYRCL - The South Yuba River Citizens League** Sponsored · 🌱 ... ✕

Help support our efforts to teach more students about rivers, streams, and wildlife in their community. Give today.

Love swimming, paddling, and hiking in the Yuba watershed?

Help support our efforts to teach more students about rivers, streams, and wildlife in their community.

Give today.



 give.yubariver.org
Love Swimming, Paddlin...
Nonprofit organization

Donate now

1 comment

 Like  Comment  Share

Treatment

 **SYRCL - The South Yuba River Citizens League** Sponsored · 🌱 ... ✕

Your \$50 donation lets us teach one additional student about rivers, streams, and wildlife in their community.

Love swimming, paddling, and hiking in the Yuba watershed?

Your \$50 donation lets us teach one additional student about rivers, streams, and wildlife in their community.

Give today.



 give.yubariver.org
Love Swimming, Paddlin...
Nonprofit organization

Donate now

 Like  Comment  Share

Control

 **yubariverpeople** Sponsored ... ✕

Love swimming, paddling, and hiking in the Yuba watershed?

Help support our efforts to teach more students about rivers, streams, and wildlife in their community.

Give today.



 Help support our efforts to teach more students about the Yuba River.



Instagram is not affiliated with this organization

Treatment

 **yubariverpeople** Sponsored ... ✕

Love swimming, paddling, and hiking in the Yuba watershed?

Your \$50 donation lets us teach one additional student about rivers, streams, and wildlife in their community.

Give today.



 \$50 lets us teach one additional student about the Yuba River.

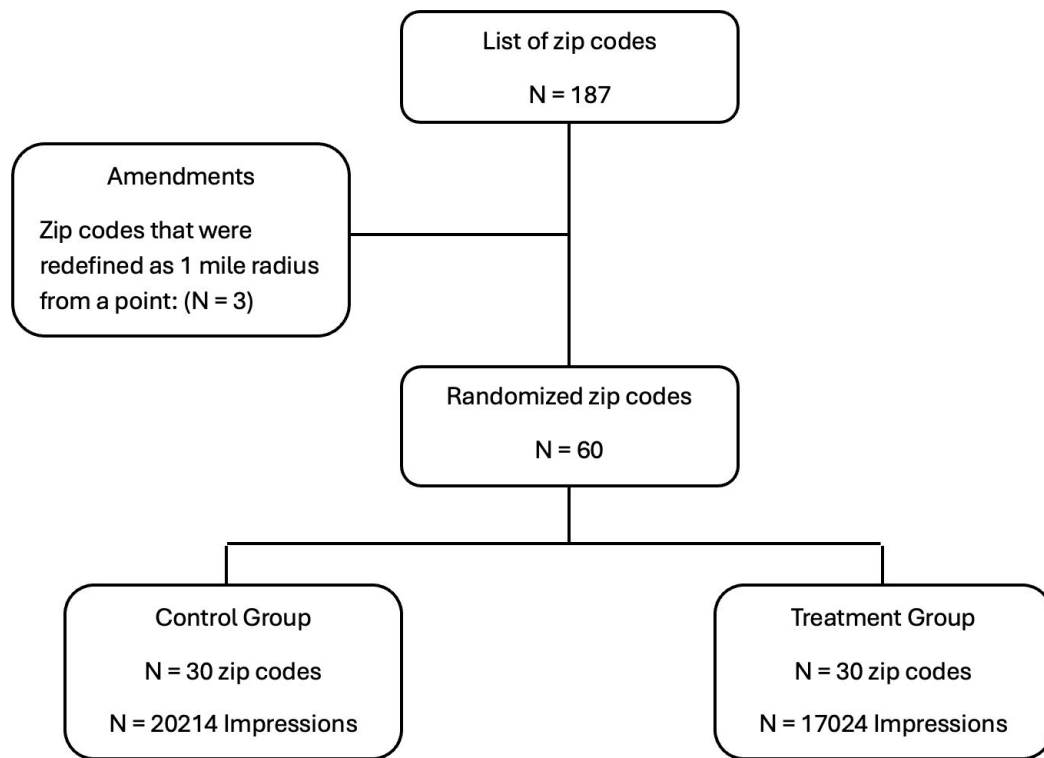


Instagram is not affiliated with this organization

Experiment Design: Measurement units

Campaigns 1 selected		Ad sets 30 selected		Ads for 30 Ad sets		Mar 1, 2025 – Mar 29, 2025				
+ Create		Duplicate		Edit		More		Columns: Performance and clicks		Breakdown
✓	Off / On	Ad set	Impressions	CPM (cost per 1,000...)	Link clicks	CPC (cost per link click)	CTR (link click-throug...)	Clicks (all)	CTR (all)	CPC (all)
✓	☒	Control 95991	938	\$7.60	4	\$1.78	0.43%	6	0.64%	\$1.19
✓	☒	Control 959...	87	\$10.23	2	\$0.45	2.30%	3	3.45%	\$0.30
✓	☒	Control 95961	893	\$6.51	7	\$0.83	0.78%	11	1.23%	\$0.53
✓	☒	Control 959...	81	\$10.86	—	—	—	—	—	—
✓	☒	Control 958...	878	\$7.72	5	\$1.36	0.57%	6	0.68%	\$1.13
✓	☒	Control 958...	924	\$6.47	3	\$1.99	0.32%	3	0.32%	\$1.99
✓	☒	Control 95815	944	\$6.26	6	\$0.99	0.64%	8	0.85%	\$0.74
Results from 3 Excludes delet...			20,215 Total	\$7.52 Per 1,000 Impressi...	110 Total	\$1.38 Per Action	0.54% Per Impressions	164 Total	0.81% Per Impressions	\$0.93 Per Click

CONSORT flow diagram

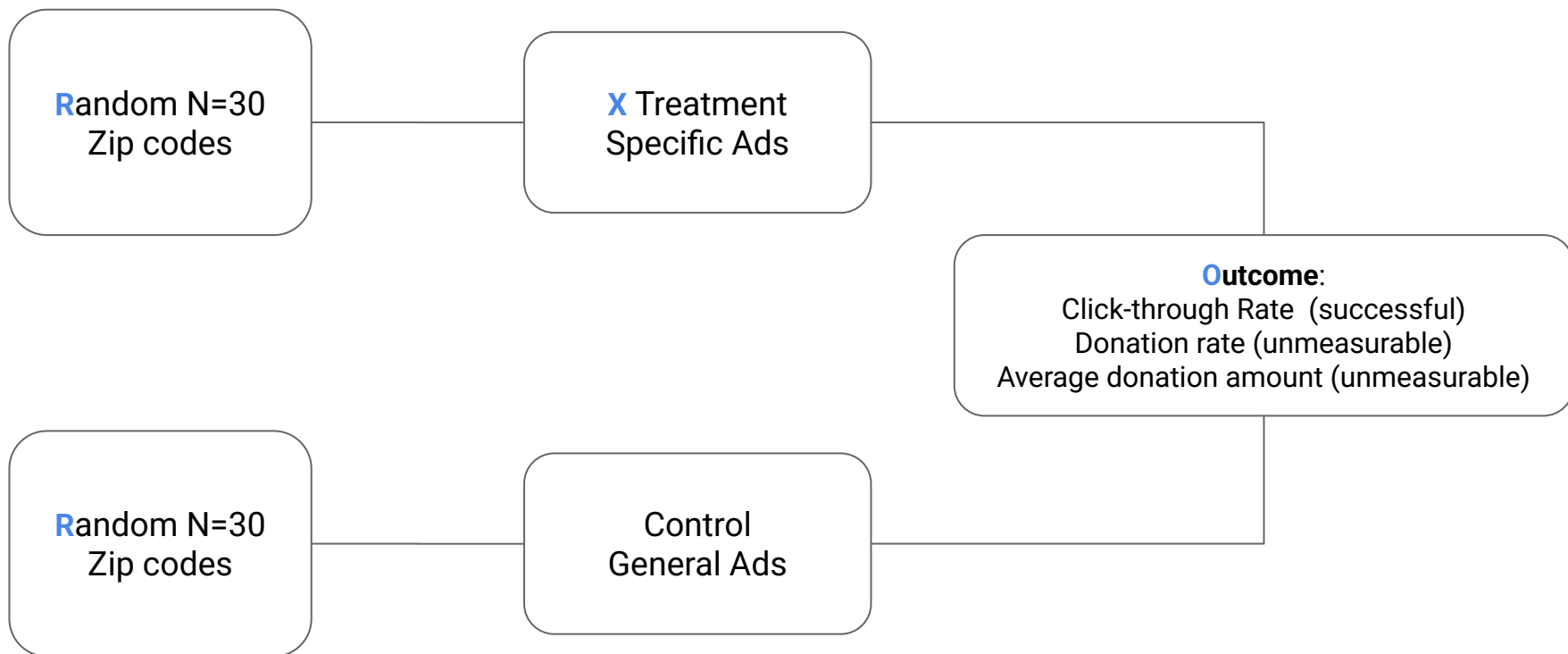


Original list of zip codes were zip codes that had previously donated to SYRCL

Top 60 zip codes that donated to SYRCL were selected, and randomized into treatment and control

Experiment design in ROXO

Between Subjects Comparison



Outcome

We measure our outcome of donation behavior by the **click-through-rate (CTR)** on the ads

- CTR best represents a donor interaction that individuals have with the ads when shown
- CTR is a common metric in advertising and marketing, because it captures an individual's interest and attention via their interaction with content

Covariate Balance Check: Gender

Table 1: Covariate Balance Check for Gender

Gender	Control_Prop	Treatment_Prop	p_value
female	0.5387	0.5359	0.5984
male	0.4487	0.4549	0.2353
unknown	0.0125	0.0091	0.0019

- Female and Male are balanced, but not for Unknown.
- Hesitated to drop Unknown rows as unsure for reason that it is unknown.
- Best to control for Gender.

Covariate Balance Check: Age

Table 2: Covariate Balance Check for Age Buckets

Age	Control_Prop	Treatment_Prop	p_value
18-24	0.1799	0.2794	0
25-34	0.1791	0.2313	0
35-44	0.1655	0.1355	0
45-54	0.1320	0.1111	0
55-64	0.1507	0.1107	0
65+	0.1928	0.1320	0

- No covariate balance for all age buckets.
- Need to control for Age.

Covariate Balance Check: Device

Table 3: Covariate Balance Check for Device Categories

Device	Control_Prop	Treatment_Prop	p_value
iphone	0.7131	0.7985	0.0000
android_smartphone	0.2525	0.1687	0.0000
ipad	0.0081	0.0085	0.7171
android_tablet	0.0018	0.0027	0.0754
desktop	0.0241	0.0212	0.0679
other	0.0004	0.0004	1.0000
ipod	0.0000	0.0001	0.4058

- No covariate balance for iPhone and Androids.
- Need to control for Device.

Covariate Balance Check: Placement

Table 4: Covariate Balance Check for Placement Categories

Placement	Control_Prop	Treatment_Prop	p_value
Instagram Stories	0.3060	0.5376	0.0000
Feed	0.1920	0.1501	0.0000
Facebook Stories	0.0506	0.0305	0.0000
In-stream video	0.3626	0.2156	0.0000
Instagram Reels	0.0441	0.0287	0.0000
Marketplace	0.0164	0.0133	0.0181
Right column	0.0226	0.0200	0.0939
Explore	0.0021	0.0014	0.1214
Facebook profile feed	0.0013	0.0014	0.9774
Search results	0.0024	0.0014	0.0258

- No covariate balance for more than half of the placements.
- Need to control for Placement.

Model Building: Overview

Model 1: $\text{CTR} \sim \text{Treatment}$

with Age and Gender covariates

Model 2: $\text{CTR} \sim \text{Treatment} +$
Age + Gender

Model with interaction
terms yielded nothing
significant

NOTE:

- *Meta Ads platform only allows us to download age and gender covariates, or device and placement covariates, but not all four at the same time.*
- *We used clustered errors on zip code, and weighted on impressions.*

with Device and Placement covariates

Model 3: $\text{CTR} \sim \text{Treatment} +$
Device + Placement

Model 4: $\text{CTR} \sim \text{Treatment}$
 $+ \text{Device} + \text{Placement} +$
Treatment:Device +
Treatment:Placement

Experiment Results: Model 1

Model 1: CTR $\sim$ Treatment

Model 1 Findings:

- Statistically insignificant coefficient for treatment assignment.

Experiment Results: Model 2 and 3

Model 2: CTR $\sim$ Treatment +
Age + Gender

Model 3: CTR $\sim$ Treatment +
Device + Placement

Model 2 & Model 3 Findings:

- Statistically insignificant coefficient for treatment assignment.
- Both models show the same treatment assignment coefficient.
 - Unexpected yet reassuring outcome, indicating no single model stands out.
- Note: Covariate imbalance remains, so coefficient may still be biased due to unobserved heterogeneity.

Experiment Results: Model 4

Model 4: CTR $\sim$ Treatment
+ Device + Placement +
Treatment:Device +
Treatment:Placement

Model 4 Findings:

- Statistically insignificant coefficient for treatment assignment.
- Users on iPads less likely to click on the treatment ad, despite balanced covariates between control and treatment groups.
- Suggests Meta's algorithm might not have accounted for this discrepancy.

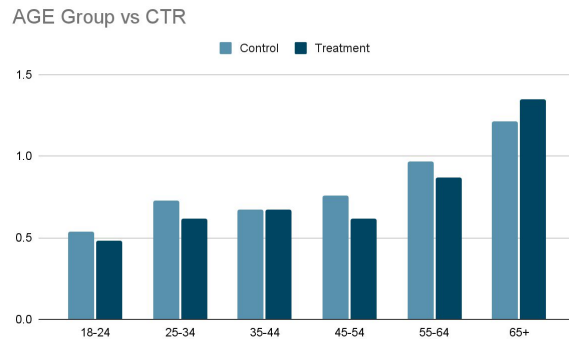
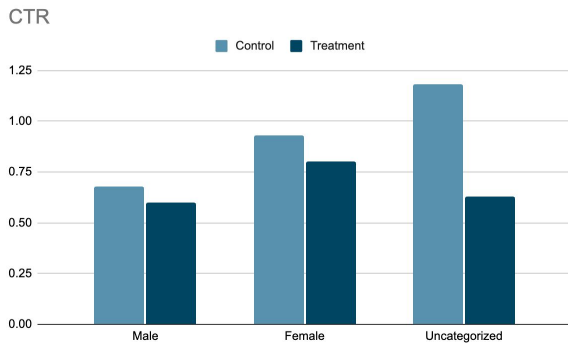
General Findings

Demographic Insights

- Users in the age groups 55-64 ($p < 0.05$) and 65+ ($p < 0.01$) are more likely to click compared to the baseline (18-24 years old).
- Male users are less likely to click relative to female users ($p < 0.01$).

Placement Performance

- All tested placements achieved a statistically significant lower click-through rate compared to the baseline of Facebook Stories.



Further Questions

1. Is there a way to mitigate the fact that we can't get all covariate data?
2. Is there a way to overcome Meta Ad's targeting algorithm so we could get better covariate balance?
3. Did SYRCL's messaging and the small text in the ad not differentiate treatment and control sufficiently?

Reference

Cryder, C. E., Lowenstein, G., & Scheines, R. (2013). The donor is in the details. Elsevier. Organizational Behavior and Human Decision Processes, Vol 120 p15-23. <https://www.cmu.edu/dietrich/sds/docs/loewenstein/DonorInDetails.pdf>